Sales Report

Performance Overview and Key Insights

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Perspective

This analysis was conducted from a sales analytics perspective, with the primary objective of evaluating revenue generation, profitability, identify top performing products and territories to support decision making . The analysis focus on four major areas;

- Sales overview - how much was sold
- Cost and profit analysis - how much it costed and how much profit was made
- Product performance - what are customers buying
- Territory analysis - where are sales coming from

Key Assumptions

The following assumptions were made to ensure consistency and analytical validity

- StandardCost was assumed to represent the actual cost of goods sold as detailed operational cost breakdown was not available
- Missing categorical values replaced with unknown were assumed not to significantly distort customer or product segmentation
- OrderDate was assumed to represent the point of revenue recognition and was used for the trend and time series analysis

Methodology

The AdventureWorks dataset, containing sales, product, customer, and HR information, was used in this project. The relevant tables required for analysis were imported into Power BI. They include; Sales.SalesOrderDetail, Sales.SalesOrderHeader, Production.Product, Production.ProductSubcategory, Production.ProductCategory, Sales.Customer, Person.Person, and Sales.SalesTerritory.

After importing the tables into Power BI, the following data transformation was done using Power Query Editor in order to prepare the data for analysis;

- Column data types were corrected where necessary. E.g. ID columns were converted from date to text format.
- Certain categorical fields that contained abbreviations were transformed into meaningful labels to improve interpretability. E.g. Product Class; L → Low, M → Medium, H → High: Product Line; M → Mountain, R → Road, S → Standard, T → Touring: Product Style; W → Women, M → Men, U → Universal.
- Columns containing null values were reviewed. Where appropriate, missing categorical values were replaced with "Unknown" and irrelevant columns with excessive null values were removed. This ensured that missing data would not distort aggregations or visual interpretations.
- After completing data cleaning, the dataset was loaded into Power BI using the Close & Apply function.
- A relational data model was constructed and relationships were established using primary and foreign keys such as: SalesOrderDetail[ProductID] → Product[ProductID], SalesOrderHeader[CustomerID] → Customer[CustomerID], SalesOrderHeader[TerritoryID] → SalesTerritory[TerritoryID], Product[ProductSubcategoryID] → ProductSubcategory[ProductSubcategoryID]. All relationships were configured as one-to-many.

- The following core measures were created: Total Revenue, Total Cost, Total Profit, Profit Margin (%), Total Orders. These measures formed the analytical foundation for all dashboard visualizations.

Each area was translated into interactive dashboard pages using appropriate visualizations such as KPI cards, line charts, bar charts, scatter plots, maps, pie and donut chart.

Key Insights

1. Overall Sales Performance

The company generated a total revenue of 109.85 million, with 275,000 units sold across 31,465 orders, serving approximately 19,820 customers. Revenue and sales volume showed consistent growth from 2011 to 2013, peaking in 2013 before declining in 2014. This indicates that the company experienced strong expansion during the early period, followed by a slowdown that may reflect market saturation, competitive pressure, or operational inefficiencies.

Universal style products account for the highest quantity sold (over 56%). This suggests that broad-appeal products drive volume. The scatter plot reveals a positive relationship between order size and revenue. It also shows a concentration of smaller orders generating moderate revenue, with fewer high quality, high revenue transactions.

2. Cost and Profit Analysis

Total cost stands at approximately 100 million, resulting in a total profit of 9.37 million and an overall profit margin of 9%. While the company is profitable, the margin indicates limited efficiency in cost control or pricing strategy. A 9% margin suggests that a significant proportion of revenue is absorbed by production and operational costs. The line charts indicate that production and operational cost increased from 2011 through 2013, before declining in 2014. However, the profit by year line chart shows a sharp increase in profit beginning in 2013 and remaining strong into 2014.

The profit margin by year line chart shows recovery after a dip in 2012, reaching its highest level in 2014. This indicates that the company became more efficient at converting revenue into profit, even as overall sales delinked. This improvement in profit demonstrates operational efficiency gains and cost management.

3. Product Performance

The Bikes category dominates revenue generation, contributing more than 86% of total revenue. Within this category, Mountain Bikes represent the most profitable subcategory. The Road and Mountain bikes are the strongest contributors to overall revenue and profit, while Standard and Touring lines generate significantly lower revenue and may not be contributing proportionally to profitability. Additionally, the top 10 products contribute a substantial share of total revenue, indicating a concentration of performance among a limited number of high-performing items.

4. Territory Analysis

The United States generates the highest revenue contribution (approximately 63 million), making it the company's primary market. The Southwest territory leads in total units sold and total revenue, while Australia generates the highest profit among territories. However, the presence of negative profit in selected territories indicates uneven performance and potential operational inefficiencies across regions.

Recommendations

1. Improve Profit Margins Through Cost Optimization

Given the current 9% profit margin, management should conduct a detailed cost review to identify inefficiencies in production, distribution, and regional operations. Pricing strategies for high-volume products such as Road bikes should be reassessed to improve margin performance without negatively affecting demand.

2. Focus on High-Performing Product Lines

The Mountain and Road product lines should receive priority investment in marketing, product development, and inventory management. Given their strong contribution to revenue and profit, expanding these lines could further increase profitability.

Conversely, underperforming lines such as Standard and Touring products should be reviewed to determine whether repositioning, cost reduction, or discontinuation is necessary.

3. Strengthen High-Value Market Segments

The United States and Southwest territories should remain strategic focus areas due to their strong revenue contribution. Best practices from profitable regions such as Australia should be analyzed and replicated in weaker territories.

For regions experiencing losses, a review of pricing, logistics costs, and customer demand patterns should be conducted to restore profitability.

4. Protect and Expand Top Products

Since a small group of products contributes significantly to revenue, inventory planning and supply chain stability for these products should be prioritized. Additional marketing efforts may further increase their contribution.

5. Address Revenue Decline After 2013

The decline in revenue observed after 2013 requires investigation. Possible factors such as market competition, pricing adjustments, customer retention, or economic conditions should be analyzed to prevent further slowdown.

Conclusion

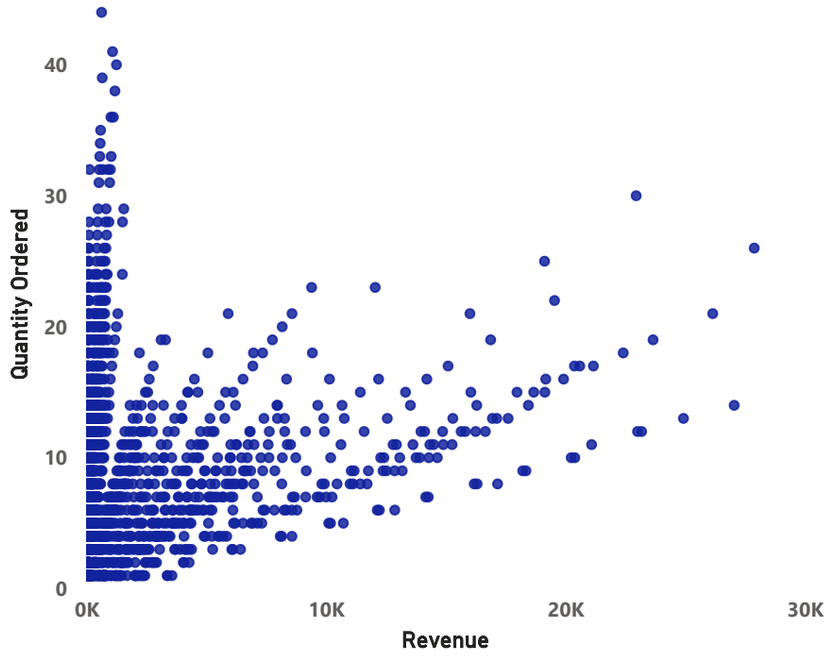
Overall, the company demonstrates strong revenue generation capacity and clear product strengths, particularly within the Bikes category and Mountain product line. However, profitability margins remain modest, and regional performance is uneven.

By focusing on cost optimization, high-performing products, profitable territories, and strategic pricing improvements, the company can enhance shareholder value, increase profitability, and support sustainable long-term growth.

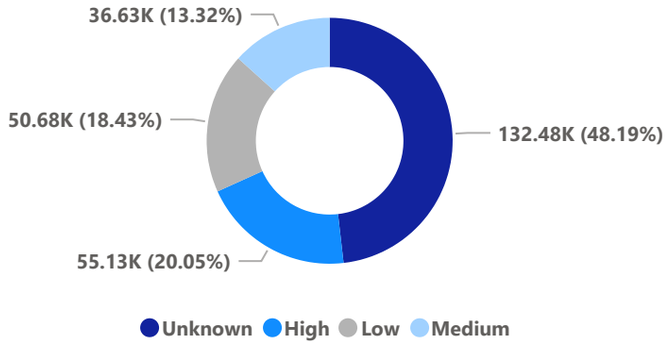
SALES OVERVIEW DASHBOARD

Total Orders	Total Quantity Sold	Total Revenue	Total Customers
31.465K	275K	109.85M	19.82K

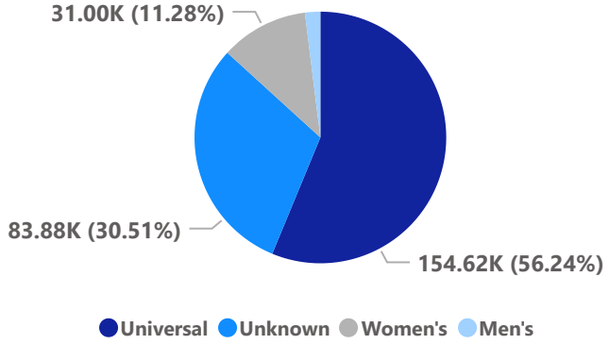
Relationship between Quantity Ordered and Revenue



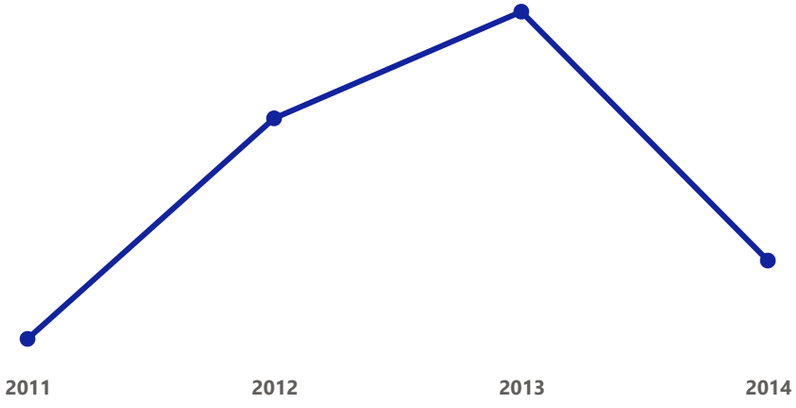
Total Quantity by Class



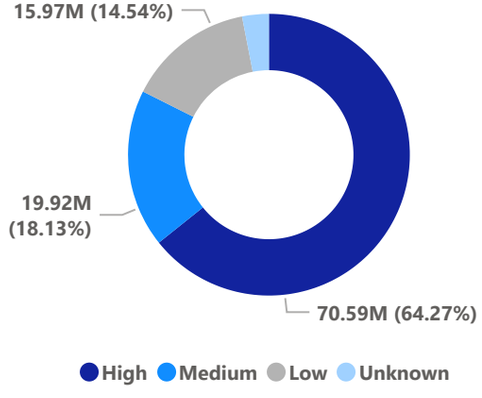
Total Quantity by Style



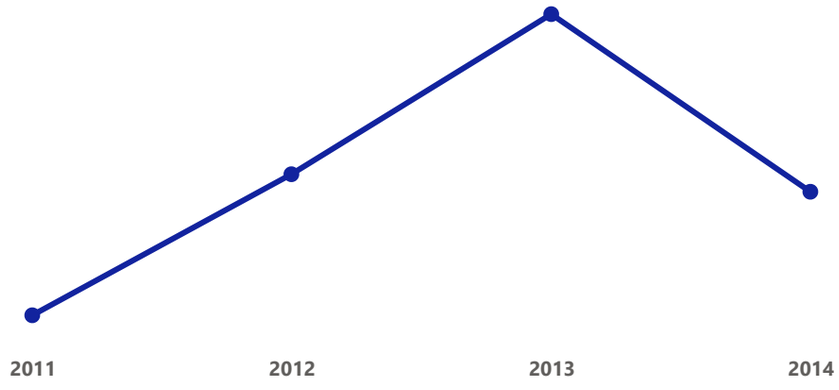
Total Revenue by Year



Total Revenue by Class



Total Quantity by Year



COST AND PROFIT ANALYSIS

Total Cost

100.47M

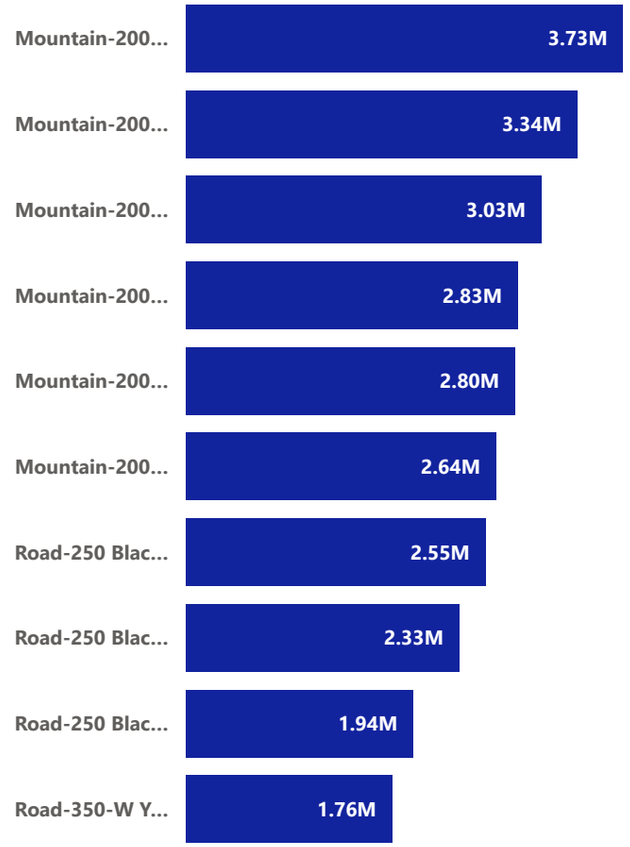
Total Profit

9.37M

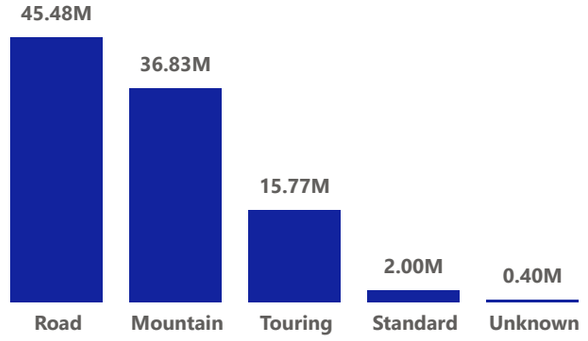
Profit Margin

0.09

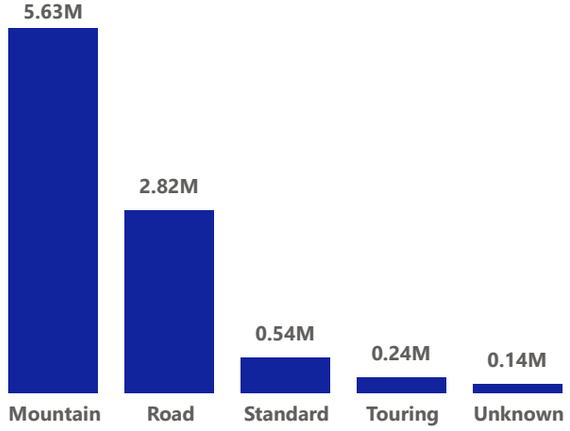
Top 10 Product by Cost



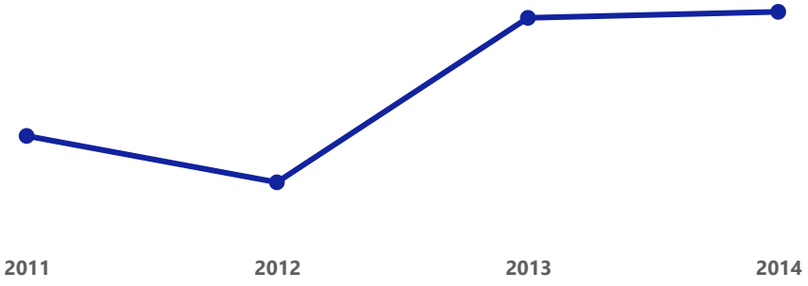
Total Cost by Product Line



Total Profit by Product Line



Total Profit by Year



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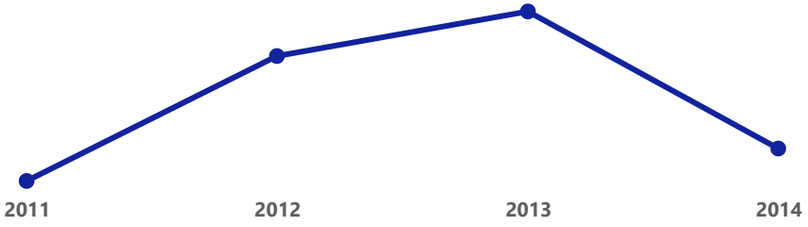
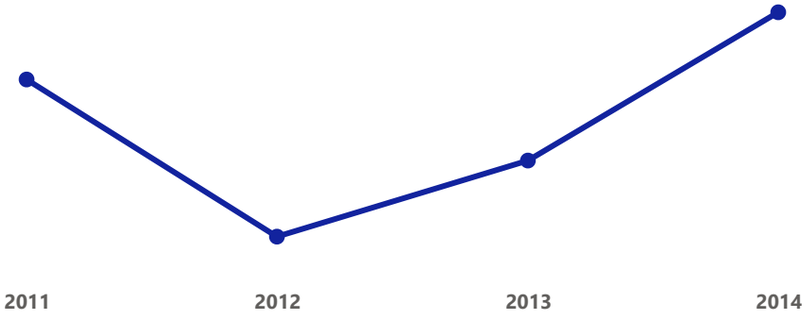
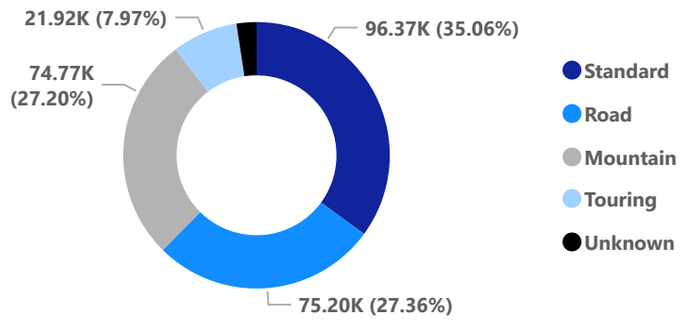


Profit Margin by Year

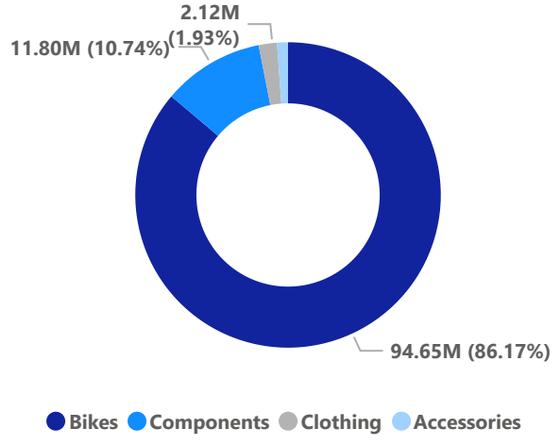


PRODUCT PERFORMANCE

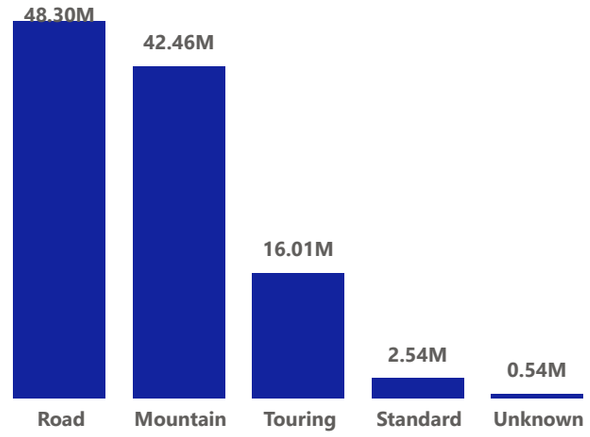
Total Quantity by Product Line



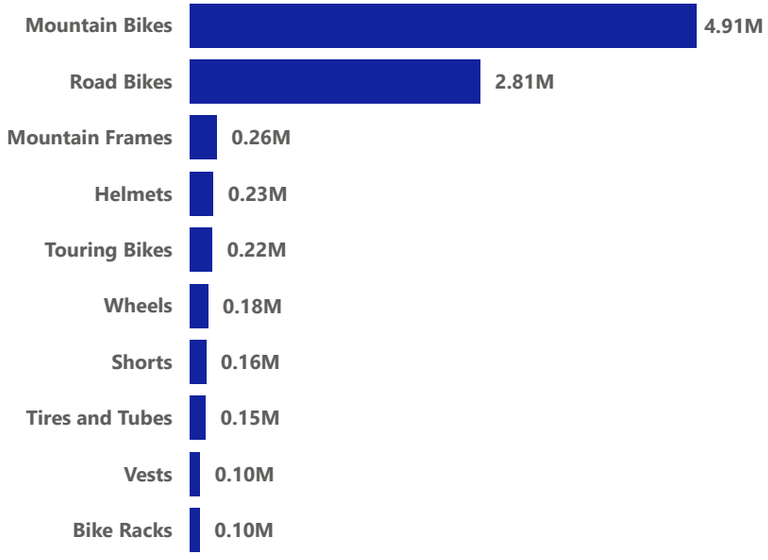
Total Revenue by Product Category



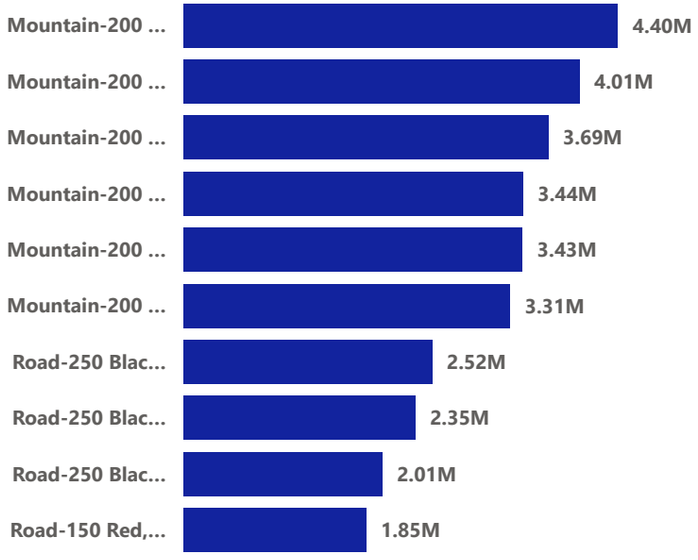
Total Revenue by Product Line



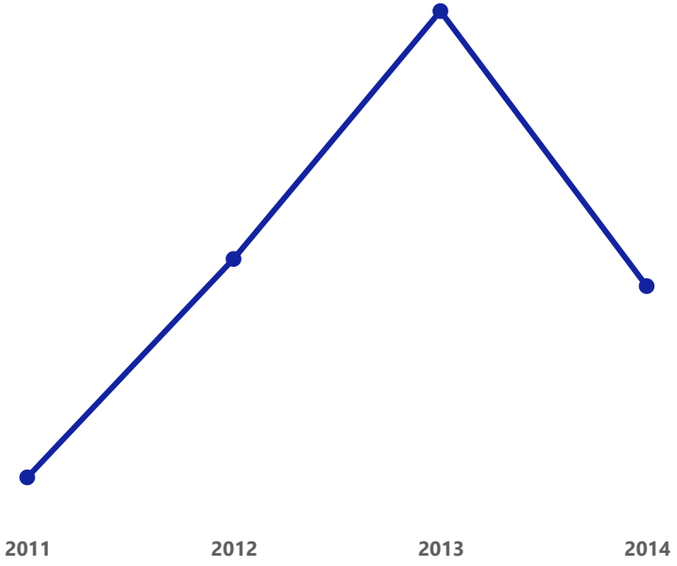
Total Profit by Product Subcategory



Top 10 Products by Revenue

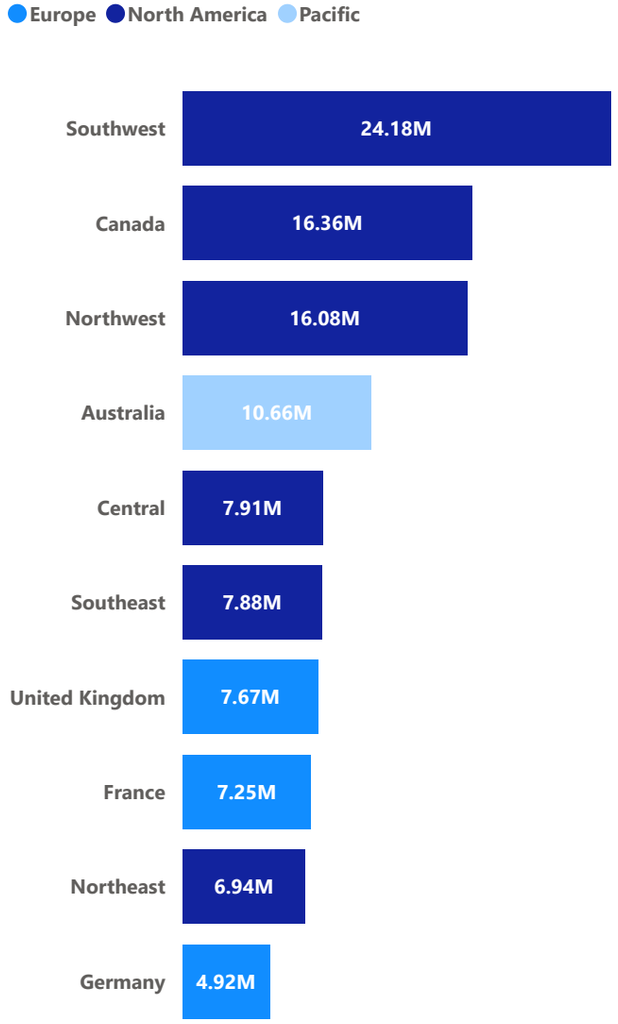


Total Product Sold by Year

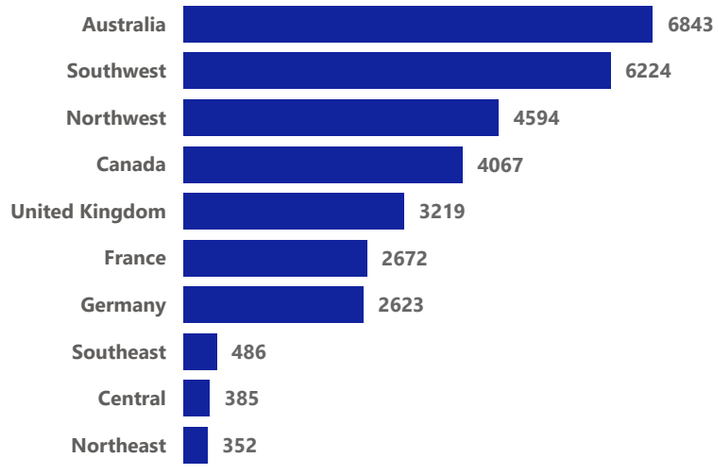


TERRITORY ANALYSIS

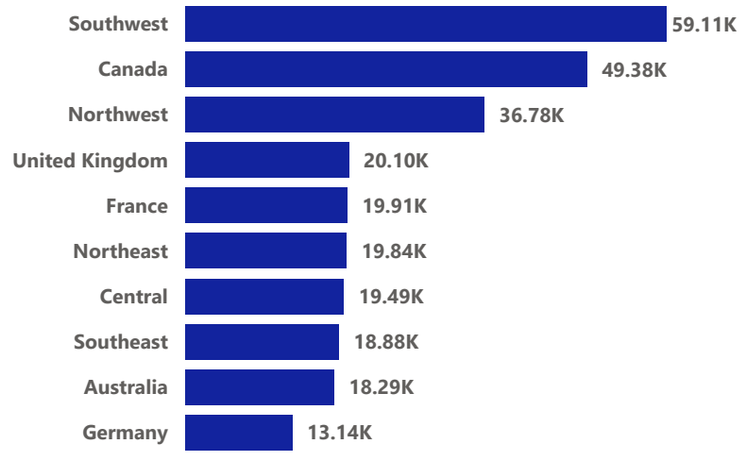
Total Revenue by Territory and Group



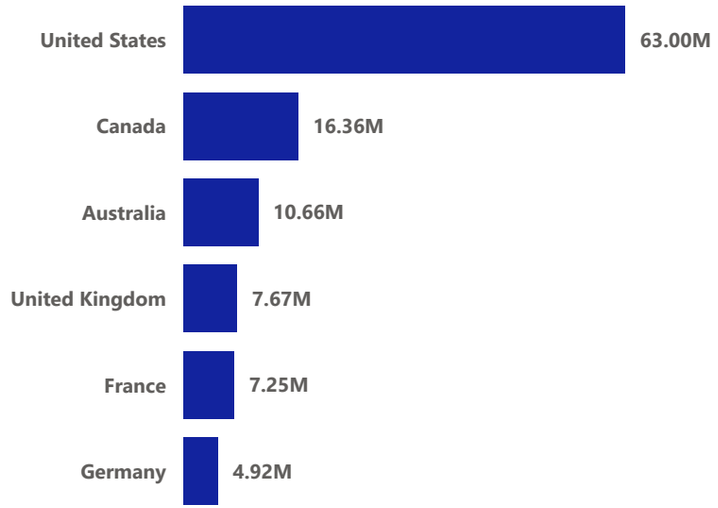
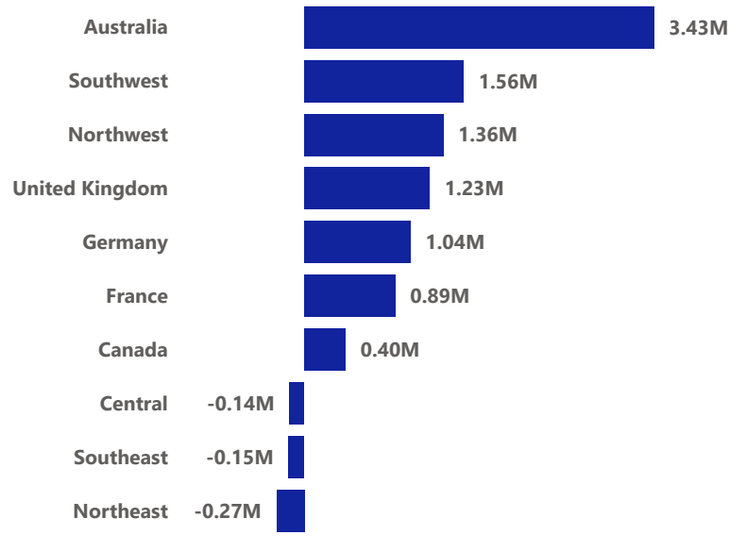
Total Orders by Territory



Total Product Sold by Territory



Total Revenue by Country



Total Profit by Territory